

# Autonomous Software Engineer & Maintainer

Codebase Mapping, Bug Investigation, Test Generation & Safe PR Creation

## Scenario Brief: Continuous Code Maintenance & Automated Bug Triage

**Company:** SaaSScale Inc. (Cloud Platform).

**Objective:** Triage incoming Sentry error reports, navigate complex codebases, write reproducing unit tests, fix the underlying bug, and submit ready-to-review Pull Requests.

### THE 5-STEP CODE ENGINEERING LOOP

- Repo Mapping (`repo_map`):** Parses AST symbols across 240+ project files in 34ms, mapping functions, types, and dependencies.
- Bug Reproduction:** Generates a standalone pytest/cargo test reproducing the exact exception in an isolated sandbox.
- Targeted Code Modification:** Edits source files precisely via `file_edit`, adhering to existing formatting.
- Automated Test Execution:** Runs test suites via shell, verifying that the failing test passes and zero regressions occur.
- Git Branch & PR Creation:** Commits changes with structured conventional commit messages and opens a GitHub PR.

### REAL EMPIRICAL CASE (FROM FATHOM TEST SUITE)

In live case study 5 (`docs/article/05-case-studies.md`), an autonomous Fathom engineer was tasked with building a complete Python CLI with MoM revenue analytics, realistic sample data, and pytest coverage.

- The agent wrote the code, ran tests, detected 1 calculation bug, edited the logic, and achieved **23/23 passing tests** completely autonomously in 74 seconds.

### DEVELOPER SUPERPOWER: ZERO FRICTION MAINTENANCE

Senior engineering talent is freed from triaging mundane bug reports, updating outdated dependencies, and writing boilerplate integration tests, allowing them to focus 100% on core product architecture.

**Developer Superpower:** Senior engineers focus on high-level system architecture while digital coworkers handle routine bugs, dependency upgrades, and test coverage.